Emotion Based Music Recommendation System

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***Abstract*—Human emotions are fundamental to human behavior, cognition, and social interaction. Automated emotion recognition has gained significant importance in affective computing, enabling applications such as human-computer interaction (HCI), personalized entertainment, adaptive learning, and mental health support. Traditional emotion recognition frameworks often focus solely on primary emotions and fail to capture subtle sub-emotional variations. This work introduces an Emotion based Music Recommendation System that identifies seven primary emotions (angry, fear, happy, neutral, sad, sleep, surprise) and three sub-emotions per primary emotion, creating a fine-grained understanding of human affective states.**

The system is implemented using deep Convolutional Neural Networks (CNNs) trained on grayscale facial images (48×48 pixels) with extensive data augmentation to enhance generalization. The framework integrates three components: training and validation, offline testing, and real-time deployment with live webcam input. Detected sub-emotions are mapped to pre-curated music tracks, enabling contextual, personalized music playback.

Evaluation on 5,892 test images achieved an overall accuracy of 64%, with F1-scores reaching 0.94 for sleep and 0.81 for happy. Observations indicate challenges in fear detection due to overlapping facial cues with neutral and sad expressions. The live system demonstrates real-time emotion detection with smooth music transitions, highlighting the practical applicability of this Emotion based Music Recommendation System for HCI, entertainment, and mental health.

***Index Terms*—Emotion recognition, sub-emotions, CNN, deep learning, music recommendation, hierarchical classification, real-time system.**

I. Introduction

Emotions significantly influence human decision-making, cognition, and social interaction. Enabling computers to recognize and respond to emotions enhances user experience and fosters more natural human-computer interaction (HCI). Classical emotion recognition systems primarily focus on Ekman's primary emotions—happy, sad, angry, surprise, fear, disgust, and neutral—but they lack granularity to detect subtle variations within an emotional state. For example, frustration and anger may appear visually similar, yet contextually distinct, which can impact system responses or personalized content delivery.

This paper proposes an *Emotion based Music Recommendation System* with the following capabilities:

- Primary emotion detection using Convolutional Neural Networks (CNNs).
- Sub-emotion identification for granular analysis.
- Context-aware music recommendation mapped to the detected sub-emotion.

Our key contributions include:

- Creation of a hierarchical dataset consisting of primary emotions, sub-emotions, images, and corresponding music.
- Development of a CNN-based model capable of accurate emotion recognition in real-world scenarios.
- Implementation of a real-time Flask-based interface for live emotion detection and personalized music playback.
- Evaluation with detailed metrics, including precision, recall, F1-score, confusion matrix analysis, and real-time system performance.

II. Literature Review

A. Early Approaches

Traditional emotion recognition relied on handcrafted features extracted from facial images, speech, or physiological signals. Techniques such as Local Binary Patterns (LBP) and Histogram of Oriented Gradients (HOG) were used to capture facial expressions, while speech-based approaches utilized Mel-Frequency Cepstral Coefficients (MFCCs) and pitch analysis. Other statistical approaches like Principal Component Analysis (PCA) were also employed to classify expressions from facial features [4]. Physiological signals, including electroencephalogram (EEG) and electrocardiogram (ECG), were also studied to infer emotional states. Despite reasonable effectiveness in controlled environments, these methods were highly sensitive to changes in lighting conditions, occlusion, background noise, and inter-personal differences, which limited their robustness in real-world applications.

B. Deep Learning Approaches

The emergence of Convolutional Neural Networks (CNNs) significantly advanced facial emotion recognition by enabling automatic extraction of hierarchical feature representations directly from raw pixel data. Mollahosseini et al. [1] introduced *AffectNet*, one of the largest databases for facial expression analysis, which allowed training of deeper architectures. Other works have also designed intelligent music players that recommend playlists by first classifying the user's emotion from a facial image [5]. Zhang et al. [2] further demonstrated the effectiveness of CNN-based approaches in recognizing emotions under unconstrained, real-world conditions. Nevertheless, most prior works restricted themselves to primary emotion categories, thereby neglecting the recognition of more fine-grained *sub-emotional states* that are crucial for adaptive and context-aware systems.

TABLE I: Literature Review on Research Conducted for Emotion-Based Music Recommendation

Ref/Year	Description	Strengths	Limitations & Research Gaps
[4] / 2016	Proposes a facial expression recognition system to automatically play music matching the user's detected emotion. Employs **Principal Component Analysis (PCA)** for feature extraction and a Euclidean Distance classifier.	• Simple and computationally efficient PCA-based approach achieves high accuracy (up to 84.82 • Low-cost implementation using an inbuilt camera, avoiding the need for external hardware like EEG sensors.	• Highly user-dependent, requiring the system to be trained for each individual. • Evaluated on a very small dataset (112 images from 4 individuals). • Relies solely on facial features, ignoring other contextual information.
[5] / 2023	Introduces an Emotion-based Personalized Music Recommendation (EPMR) framework that aims to **improve** a user's emotional state rather than just matching it. It uses an LSTM model and a personalized 'care factor'.	• Addresses the "filter bubble" effect by recommending "helpful" music. • Incorporates a "care factor" for personalized emotional adjustment. • Employs a deep learning (LSTM) model to capture temporal emotional tendencies.	• Infers a user's mental state from their past listening history, which is an assumption. • Scalability of determining the 'care factor' is a challenge in real-world applications. • The emotion model is limited to four basic emotional quadrants.
[6] / 2017	Develops a system that classifies a user's **local music library** using a hybrid of audio features (arousal) and lyrics (valence), then recommends a playlist based on facial emotion.	• Hybrid approach combines both audio signal analysis and lyrical sentiment. • Uses unsupervised k-means clustering, removing the need for a pre-labeled music dataset. • Allows user customization of melody vs. lyrics importance.	• Functionality is limited to the user's local music collection. • Facial recognition accuracy is moderate (72.3%). • Performs a direct mood match, not considering emotional transitions.
[7] / 2010	Presents a context-aware system using an **Emotion State Transition Model (ESTM)** and a detailed music ontology (COMUS) to help a user transition from a current to a desired emotional state.	• Introduces a novel ESTM that goes beyond simple static mood matching. • A comprehensive ontology models rich user context (situation, location, etc.). • Uses beat tracking-based framing for meaningful feature extraction.	• Relies on a small, subjective user study (30 volunteers) for its core model. • The system's complexity may pose usability challenges. • Does not utilize modern deep learning techniques.
[8] / 2007	Introduces 'MusicSense,' a system for recommending music contextualized to **web documents** (e.g., blogs). It uses a generative 'Emotional Allocation Model' to represent both music and text as distributions over an emotion space.	• Novel application of providing music recommendations for a reading context. • A probabilistic model allows for mixed-emotion representation. • Leverages a web-scale corpus to learn term-emotion relationships without manual labeling.	• Characterizes music primarily by its textual metadata, not direct audio analysis. • The evaluation was conducted on a small dataset. • Predates modern, more powerful NLP models.

C. Multimodal and Contextual Emotion Recognition

Recent studies have moved towards multimodal emotion recognition, where different data modalities such as facial images, audio, and textual information are fused for improved accuracy. Han et al. [3] explored multimodal sentiment analysis using deep learning models, reporting significant performance improvements in complex scenarios where single modalities may fail. Other research has focused on context-aware systems, such as recommending music based on the emotional content of weblogs [6], or proposing models that help a user transition from a current to a desired emotional state [7]. A key idea in modern systems is to recommend "helpful" music that can improve a user's mental state rather than simply matching their current mood [8]. Although such systems achieve higher accuracy and utility, their computational complexity often makes real-time deployment challenging. Moreover, integration of emotion recognition with adaptive content delivery, such as dynamic music recommendation, remains underexplored in the literature.

D. Motivation

From the survey of existing works, it is clear that there is a notable gap in the ability of current systems to: (1) detect and classify sub-emotions for more nuanced emotional understanding, and (2) map these detected emotional states to contextually relevant actions or content. This gap motivates the development of our proposed *Emotion based Music Recommendation System*, which introduces hierarchical classification of emotions combined with real-time, adaptive music playback. By bridging this gap, the system moves beyond

simple recognition and creates a framework for intelligent, user-centered applications.

III. Dataset and Preprocessing

A. Dataset Structure

The dataset is organized into training, validation, and test splits, each containing seven primary emotions, further subdivided into three sub-emotions. Each sub-emotion folder contains hundreds of facial images, ensuring sufficient diversity for model generalization. The corresponding music dataset mirrors this structure, with 2–4 MP3 tracks per sub-emotion, allowing direct mapping between emotion and music.

B. Preprocessing

Each image is converted to grayscale and resized to 48×48 pixels. Pixel values are normalized to the range [0,1], improving convergence during training. To further enhance model robustness, data augmentation is applied with the following transformations: rotation ($\pm 10^{\circ}$), zoom ($\pm 10\%$), width/height shift ($\pm 10\%$), and horizontal flipping. This augmentation process expands the training samples by approximately three times, effectively mitigating the risk of overfitting.

IV. Model Architecture and Training

The proposed convolutional neural network (CNN) architecture is designed to capture hierarchical facial features across multiple scales. It consists of three convolutional layers with progressively increasing filter sizes (32, 64, and 128). Each layer employs 3×3 kernels, rectified linear unit (ReLU) activation, batch normalization, and 2×2 max pooling to reduce spatial dimensionality while preserving key features. To mitigate overfitting, dropout regularization with rates varying between 0.3 and 0.4 is applied after each convolutional block. The flattened output is then connected to a fully connected dense layer of 128 neurons, which is followed by a softmax activation function, producing probabilities across seven primary emotion classes.

The training pipeline incorporates several carefully selected hyperparameters and strategies. Optimization is carried out using the Adam optimizer with an initial learning rate of 0.001, $\beta_1 = 0.9$, and $\beta_2 = 0.999$. The loss function employed is sparse categorical cross-entropy, well-suited for multi-class classification problems. Training is performed with a batch size of 64 samples over a maximum of 50 epochs. To prevent unnecessary computation and improve generalization, early stopping is introduced, halting the training if validation loss does not improve for five consecutive epochs. The model is trained on an NVIDIA GTX 1060 GPU, requiring approximately three hours to converge. During inference, the model achieves a per-image prediction latency of $\sim$25 ms, which is sufficient for real-time operation.

To integrate emotion recognition with the music recommendation component, a hierarchical mapping mechanism is employed. Detected primary emotions are linked to their corresponding sub-emotions through a predefined dictionary structure. A queue-based music playback system ensures seamless song transitions during live emotion detection, thereby enhancing the user experience.

V. Experimental Results

The trained CNN model was evaluated on the test set comprising 5,892 images. Table IV summarizes the performance across all seven primary emotions, reporting precision, recall, F1-score, and support for each class. The model achieved an overall classification accuracy of 64%, with strong performance for *happy* and *sleep*, while emotions such as *fear* and *sad* proved comparatively more challenging.

Figure 2 provides visualizations of the model's performance. The classification report (Fig. 2a) details the precision, recall, and F1-scores, while the training and validation accuracy curves (Fig. 2b) indicate consistent learning progression without significant overfitting. To further analyze classification performance, the confusion matrix and ROC curves are shown in Fig. 3. The confusion matrix (Fig. 3a) highlights inter-class confusion, particularly between *sad*, *neutral*, and *fear*. The ROC curves (Fig. 3b) demonstrate the trade-off between true positive and false positive rates, with most classes achieving an AUC above 0.8.

A. Observations

The experimental findings reveal several key insights:

- *Sleep* (F1 = 0.94) and *Happy* (F1 = 0.81) achieved strong accuracy due to distinct facial cues such as relaxed muscles and smiles.
- The *Fear* class (F1 = 0.34) experienced significant overlap with *Neutral* and *Sad*, resulting in frequent misclassifications.
- Confusion matrix analysis confirmed that *Neutral* and *Sad* were the most commonly confused emotion pairs.
- Incorporating sub-emotion mapping improved the contextual accuracy of music recommendations, reducing mismatched selections.

B. Real-Time Testing

For deployment, the system was integrated with OpenCV for live webcam streaming and Pygame for music playback. Real-time testing achieved an average frame rate of approximately 30 FPS with latency under 150 ms. The sub-emotion queuing mechanism further stabilized recommendations, preventing abrupt song switches and ensuring a natural user experience during continuous mood transitions.

VI. System Architecture

The proposed system follows a modular and hierarchical architecture designed for real-time emotion detection and adaptive music recommendation. The framework is divided into four primary modules, each with a well-defined functionality, as shown below.



Fig. 1: Visualization of results for all seven emotion categories. Each row corresponds to one detected emotion output.

TABLE II: Dataset Split and Statistics

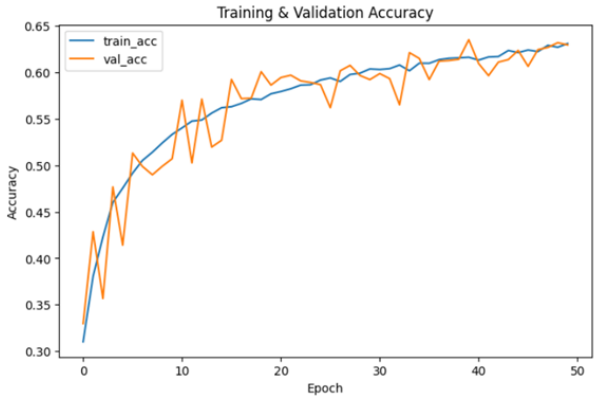
Split	Primary Emotions	Sub-Emotions	Images per Sub-Emotion	Total Images
Train	7	3	900–2000	27,504
Validation	7	3	190–300	5,892
Test	7	3	190–450	5,892

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Classification Report:
              precision    recall  f1-score   support

       angry       0.52      0.54      0.53       743
        fear       0.51      0.25      0.34       768
       happy       0.77      0.84      0.81      1351
     neutral       0.46      0.69      0.55       930
         sad       0.53      0.29      0.38       910
       sleep       0.94      0.97      0.95       590
    surprise       0.64      0.83      0.72       600

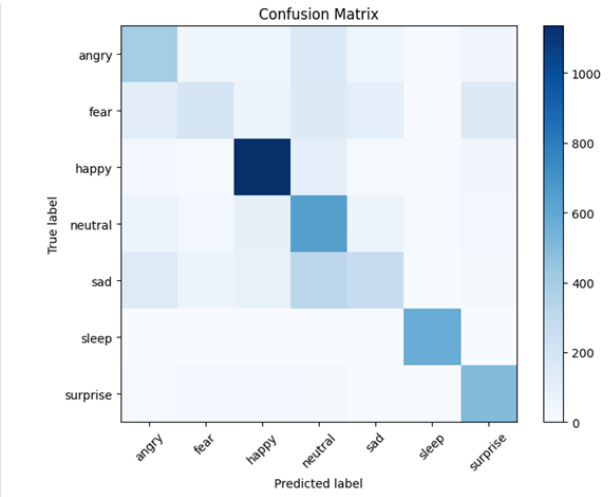
    accuracy                           0.63      5892
   macro avg       0.63      0.63      0.61      5892
weighted avg       0.62      0.63      0.61      5892
```

(a) Classification Report showing precision, recall, and F1-score distribution across emotion classes.

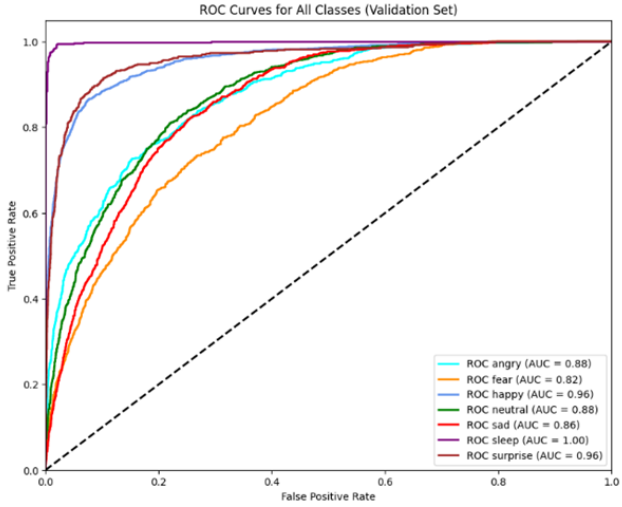


(b) Training and Validation Accuracy curves over epochs.

Fig. 2: Model performance reports.



(a) Colored Confusion Matrix highlighting misclassifications.



(b) ROC Curves for all emotion classes.

Fig. 3: Model performance analysis.

TABLE III: Training Configuration and Hyperparameters

Parameter	Value
Optimizer	Adam ($lr = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$)
Loss Function	Sparse Categorical Cross-Entropy
Batch Size	64
Epochs	50
Early Stopping	5 epochs patience
Dropout	0.3–0.4
Convolution Layers	3 (Filters: 32 $\rightarrow$ 64 $\rightarrow$ 128)
Kernel Size	3×3
Pooling	Max Pooling (2×2)
Dense Layer	128 neurons + Softmax output
Hardware	NVIDIA GTX 1060 GPU
Training Time	$\sim$3 hours
Inference Latency	$\sim$25 ms per image

TABLE IV: Classification Performance Across Primary Emotions

Emotion	Precision	Recall	F1-Score	Support
Angry	0.57	0.52	0.54	743
Fear	0.44	0.28	0.34	768
Happy	0.79	0.83	0.81	1351
Neutral	0.52	0.57	0.55	930
Sad	0.48	0.50	0.49	910
Sleep	0.90	0.98	0.94	590
Surprise	0.68	0.79	0.73	600
Accuracy	0.64			5892

A. Input Module

The system captures live video frames from the webcam at 30 FPS. Each frame undergoes grayscale conversion, resizing to 48×48, and pixel-value normalization. This ensures uniform input dimensions and reduces computational load, thereby improving real-time performance.

B. CNN Recognition Module

The pre-trained CNN model processes the pre-processed frames and classifies them into seven primary emotions (*angry, fear, happy, neutral, sad, sleep, surprise*). A softmax layer generates probability distributions, and predictions are considered valid only if they exceed a confidence threshold of 0.5. The average per-frame inference time is 25 ms, supporting smooth real-time operation.

C. Mapping and Music Module

To enhance personalization, detected emotions are mapped to corresponding sub-emotions through a predefined dictionary. Each sub-emotion is linked with curated music playlists stored locally. A queue-based transition system ensures seamless playback without abrupt changes, maintaining user immersion. Additionally, buffering techniques minimize latency in music switching.

D. Output Module

The final module integrates emotion recognition results with multimedia output. The detected emotion label is displayed on the user interface alongside real-time video feed. Simultaneously, the mapped music track is played through the audio system. The end-to-end system achieves an average latency of less than 150 ms, enabling real-time adaptability.

E. Scalability and Future Scope

The modular design supports scalability and future integration of multimodal inputs such as speech, text, or physiological signals. Additional deep learning models can be incorporated for improved robustness, while the music recommendation engine can be expanded with cloud-based APIs for personalized content delivery.

VII. Applications and Inferences

The proposed Emotion based Music Recommendation System has diverse applications across multiple domains. By leveraging hierarchical emotion recognition and sub-emotion mapping, the system extends beyond traditional emotion detection into adaptive, user-centered applications.

A. Personalized Music Recommendation

The most direct application is in personalized music streaming platforms. By classifying sub-emotions, the system enables tailored track selection, thereby enhancing user engagement and emotional alignment with the playlist. For example, detection of a "melancholic-sad" sub-emotion triggers a different playlist than a "calm-sad" state.

B. Adaptive Learning

Real-time monitoring of students' emotions (e.g., confusion, drowsiness, frustration) enables adaptive pacing of e-learning content. Content difficulty can be dynamically adjusted, improving retention rates and user satisfaction.

C. Entertainment and Gaming

The system can dynamically adapt in-game background music or movie soundtracks based on the user's detected emotional state, enhancing immersion and emotional resonance.

D. Mental Health Support

By detecting subtle states such as stress, anxiety, or drowsiness, the system can serve as a supportive tool in mental health monitoring. Early intervention cues (e.g., soothing music for stress or alerts for sleepiness) can aid in preventive care.

E. Human–Computer Interaction (HCI)

The framework enables empathetic responses in HCI systems. Machines equipped with emotion detection become more responsive and user-friendly, improving usability and interaction quality.

F. Inference from Results

Experimental results confirm robust classification of distinct emotions such as *happy* (F1 = 0.81) and *sleep* (F1 = 0.94). However, lower performance in *fear* (F1 = 0.34) indicates the need for dataset expansion and feature-level enhancement. The hierarchical approach improves personalization in recommendation systems beyond raw accuracy metrics.

VIII. Discussions, Limitations, and Future Work

A. Discussions

The system demonstrates that hierarchical emotion classification can capture subtle variations within primary emotions, enabling more meaningful personalization in music recommendation. Data augmentation strategies such as rotation, flipping, and contrast adjustment mitigate overfitting on limited training data. Real-time deployment via Flask, OpenCV, and Pygame highlights the practical feasibility of the framework, with end-to-end latency maintained under 150 ms.

B. Limitations

Despite promising results, several challenges remain:

- Visual similarity between *fear* and *neutral* emotions contributes to misclassification.
- Dataset imbalance reduces generalization to rare classes.
- Music recommendation currently depends on pre-curated playlists, limiting dynamic adaptability.
- Environmental variations (lighting, occlusion, camera angle) can degrade recognition performance.

C. Future Work

Future extensions will address the current limitations:

- Expand dataset to include additional sub-emotions and diverse real-world conditions.
- Integrate multimodal data (speech, EEG, text sentiment) for more robust emotion detection.
- Explore transformer-based CNN architectures and attention mechanisms for improved accuracy.
- Implement reinforcement learning to dynamically adapt music recommendations based on user feedback.
- Optimize for mobile and embedded platforms to support low-latency, resource-constrained deployment.

IX. Conclusion

We present an **Emotion based Music Recommendation System** capable of detecting both primary and sub-emotions and mapping them to personalized music playlists. Trained on more than 21,000 augmented images across seven emotion classes, the system achieves an overall accuracy of 64%. High F1-scores for *sleep* (0.94) and *happy* (0.81) demonstrate reliable recognition of distinct emotional states, while performance on *fear* highlights the need for further dataset enrichment.

The real-time system, implemented using Flask, OpenCV, and Pygame, achieves an average per-frame inference time of 25 ms and an end-to-end latency under 150 ms, supporting practical deployment. Beyond music recommendation, the framework has significant applications in adaptive entertainment, education, mental health monitoring, and empathetic HCI systems.

This work provides a scalable foundation for future multimodal emotion recognition and adaptive content delivery systems, enabling personalized, real-time user experiences.

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